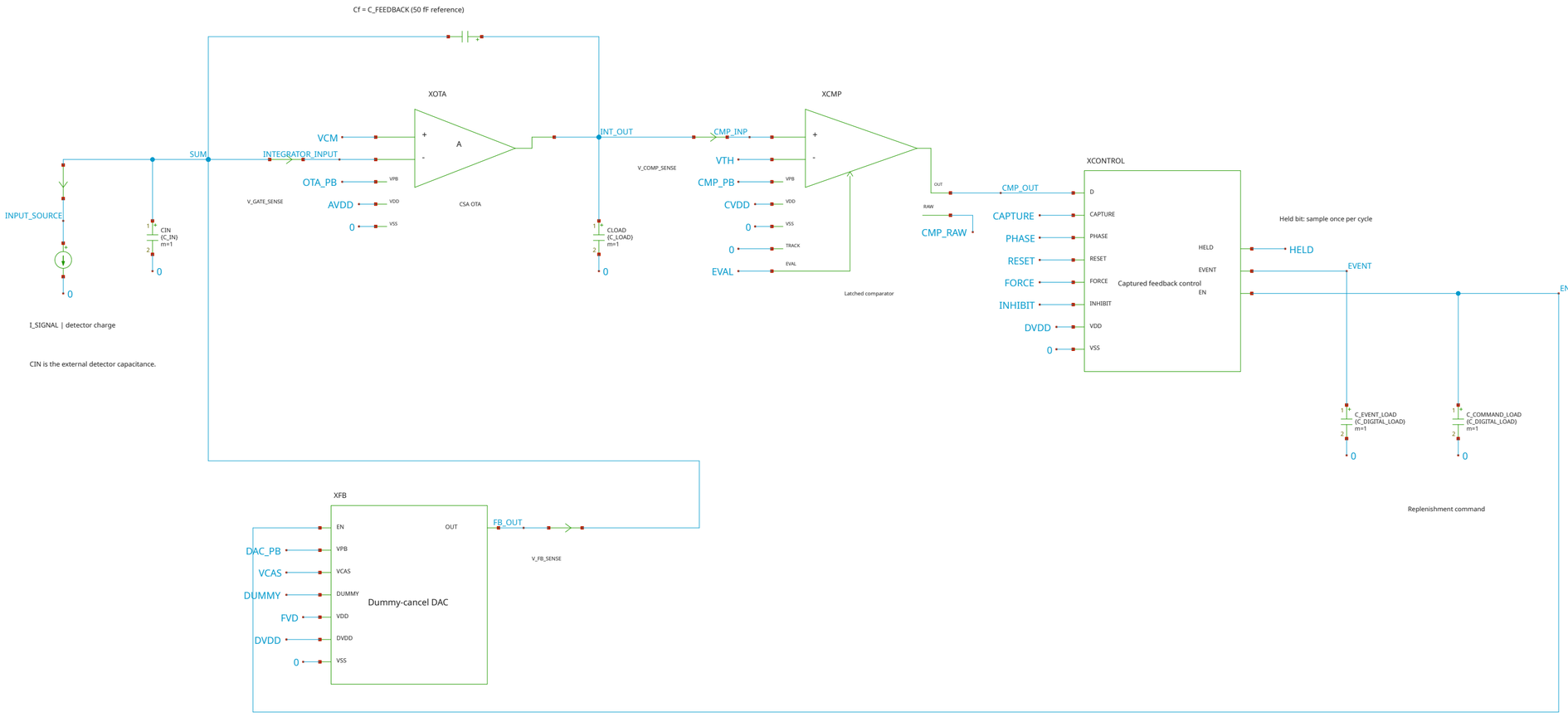
External TSMC180 model selector
Physical temperature = 298.15 K
Corner = TT

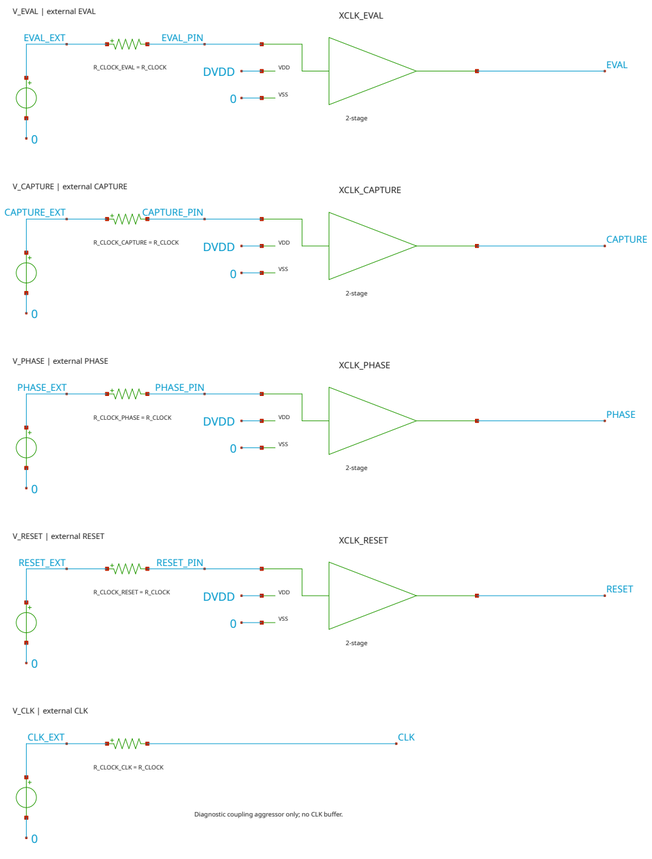
Q-Pix single-channel loop | selected preamplified architecture

Negative detector charge -> reset-free CSA -> comparator -> held EVENT -> calibrated current replenishment. Arrow symbols are ideal current probes.

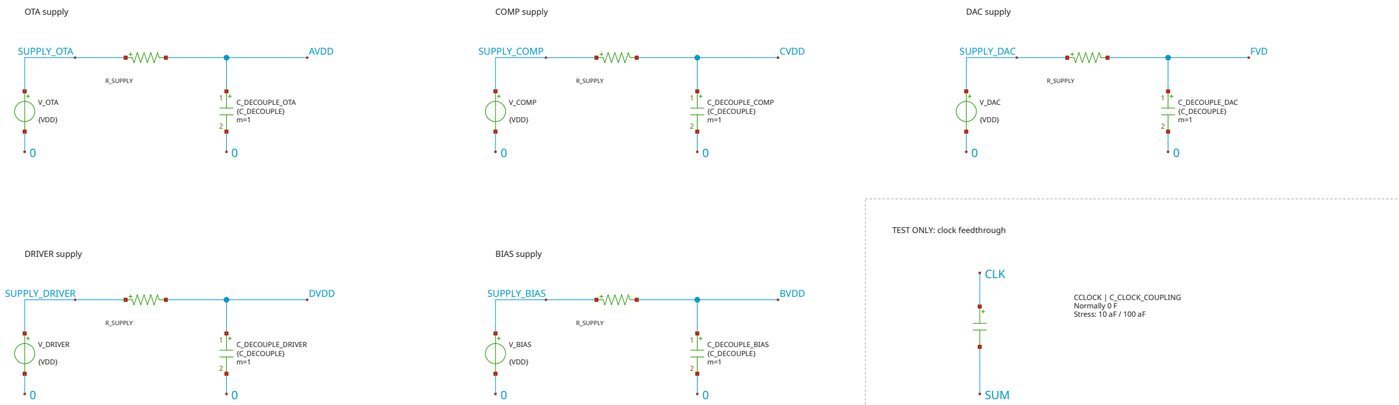


Opposing current quantum; no reset switch or resistor across Cf.

EXTERNAL TIMING + IMPLEMENTED LOCAL BUFFERS

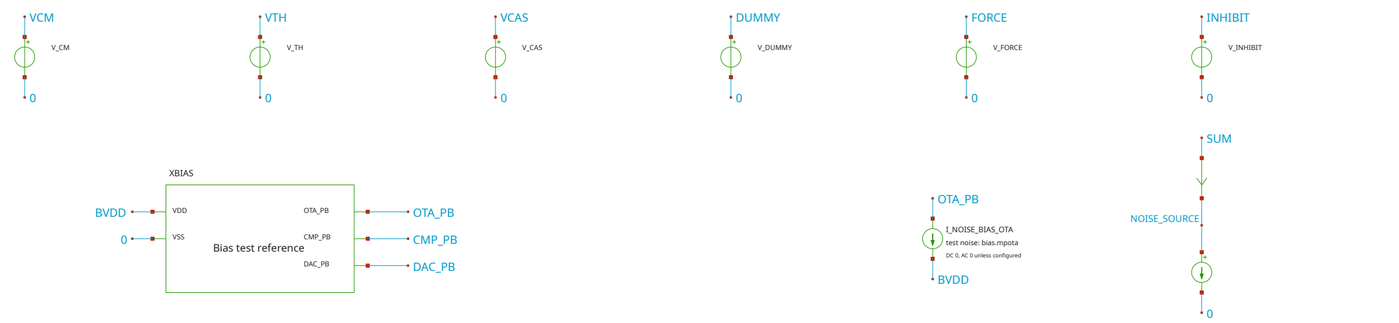


FIVE SOURCE-POWER DOMAINS | SERIES FEED + DECOUPLING





IDEAL EXTERNAL REFERENCES / TEST CONTROLS



Diode bias interfaces; external current seeds remain explicit.

Default-off equivalent noise stimulus

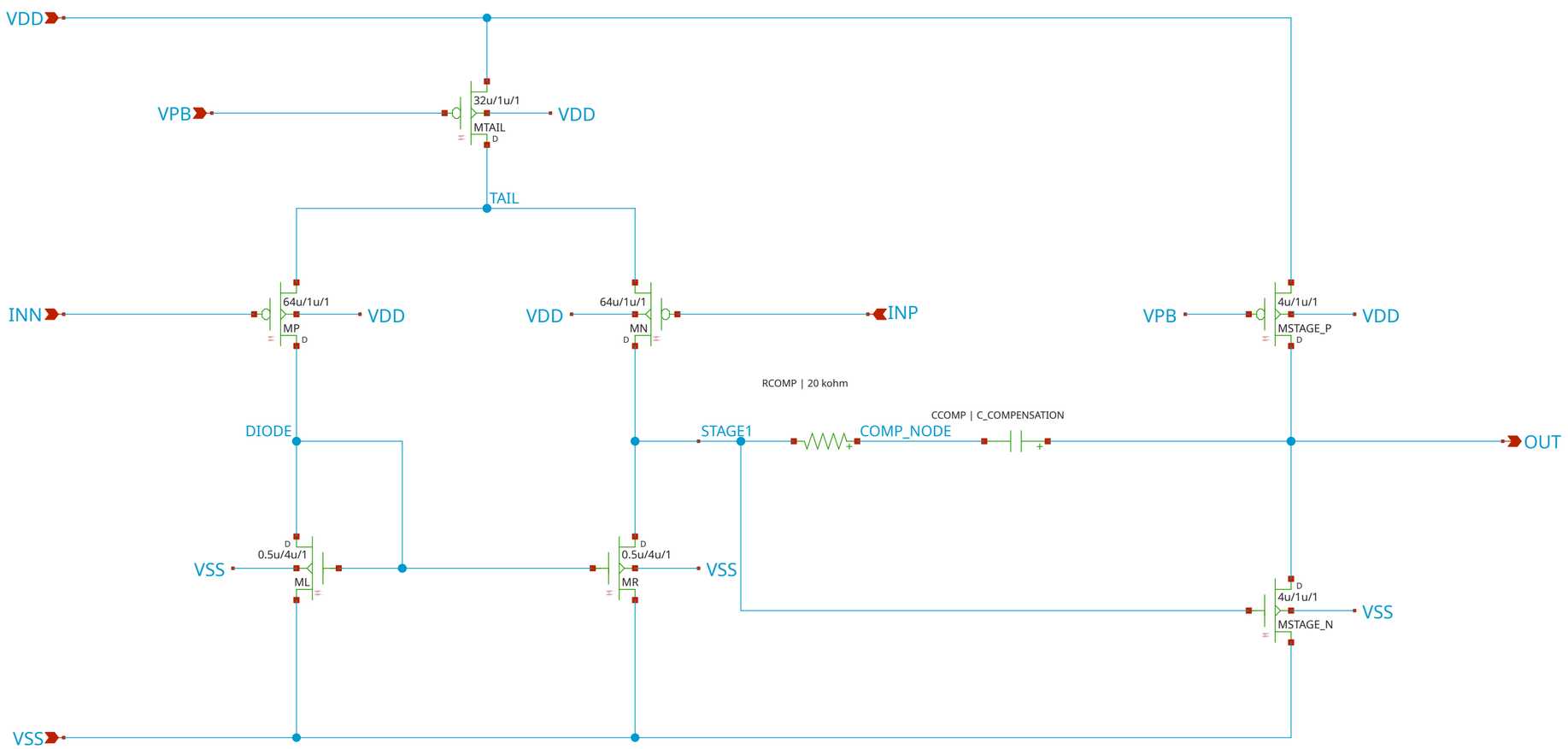
PARAMS | select the code block and press q for the complete unchanged simulator defaults.

GUI defaults: 1 MHz; C_IN=3 pF; Cf=50 fF; 1.8 V. Qualification recipes supply frozen trims and the selected late phases.

Every source, current probe, receiver load and power-feed element is retained. Colored net names connect support interfaces across panels.

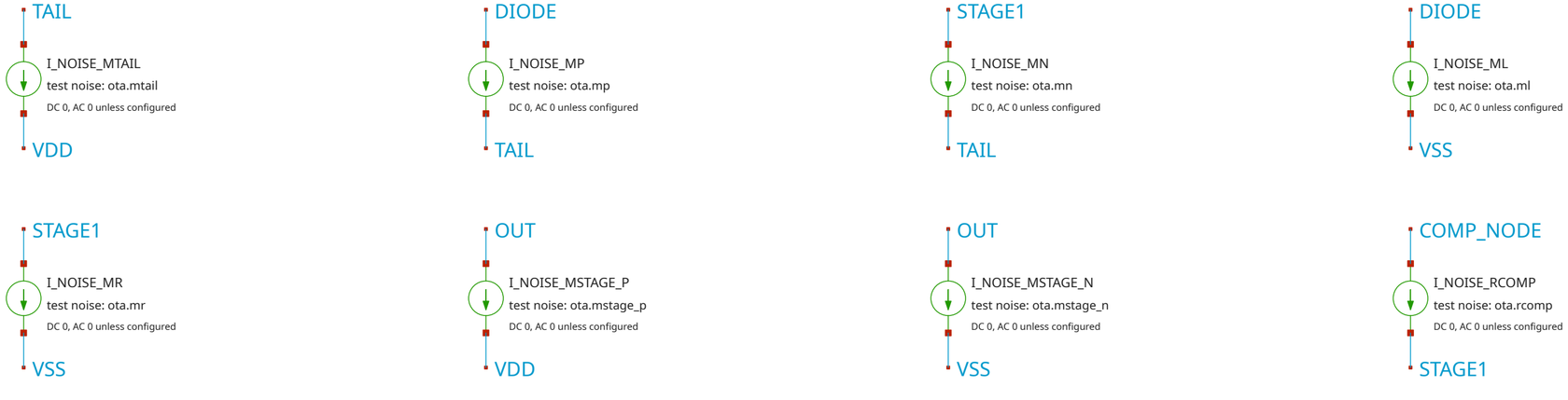
PMOS-input CSA OTA | selected front-bias allocation

Differential pair and mirror load at left; second gain stage at right; series Miller compensation between stages.



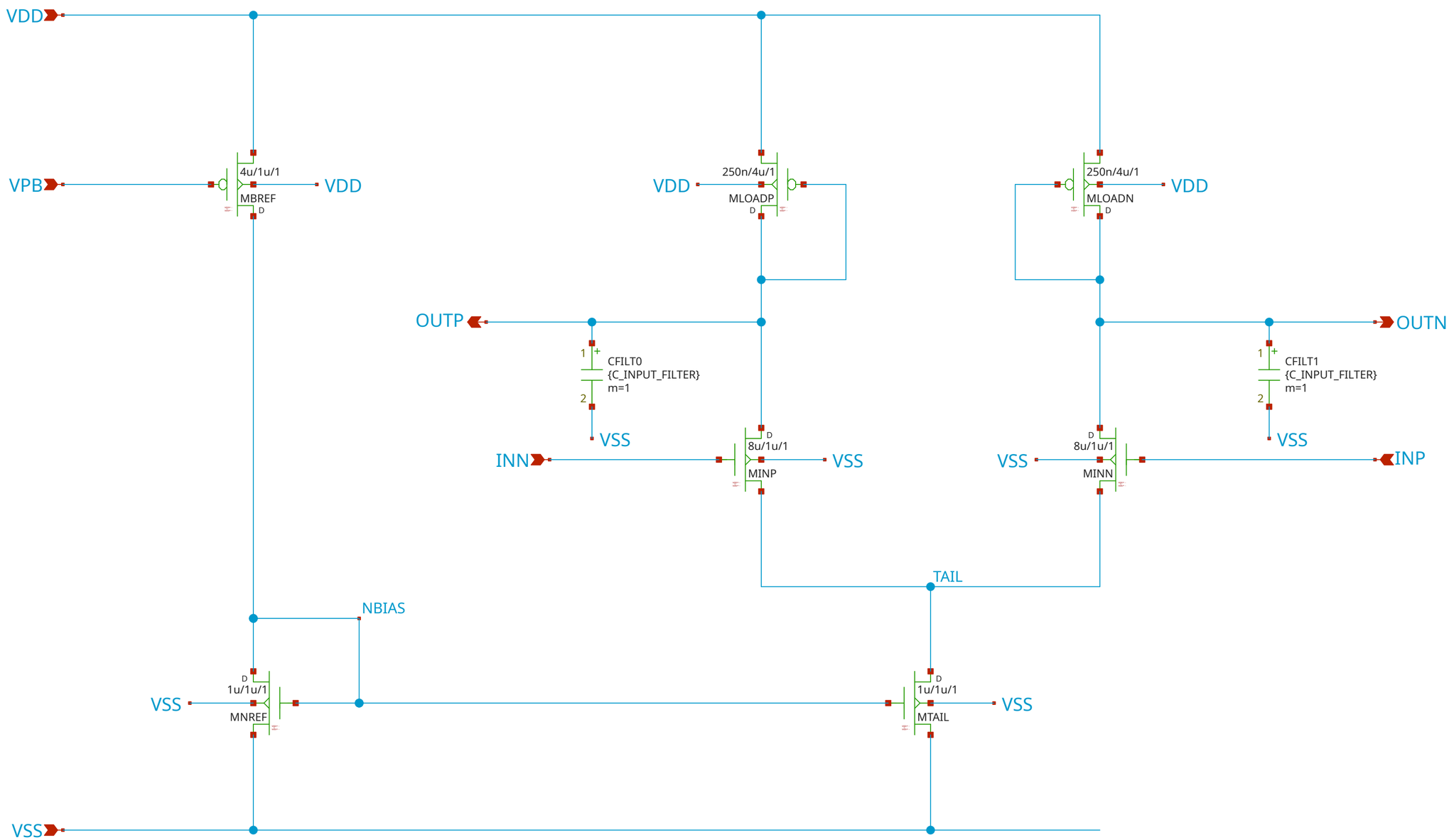
7 MOS; editable W/L and matched geometry recipe. Bulk ties use supply labels.

TEST-ONLY NOISE PORTS | default DC/AC zero; excluded from silicon device count



Continuous differential preamplifier

NMOS input pair with diode-connected PMOS loads, explicit output filters and mirrored tail bias. No ideal gain or clocked input switch.

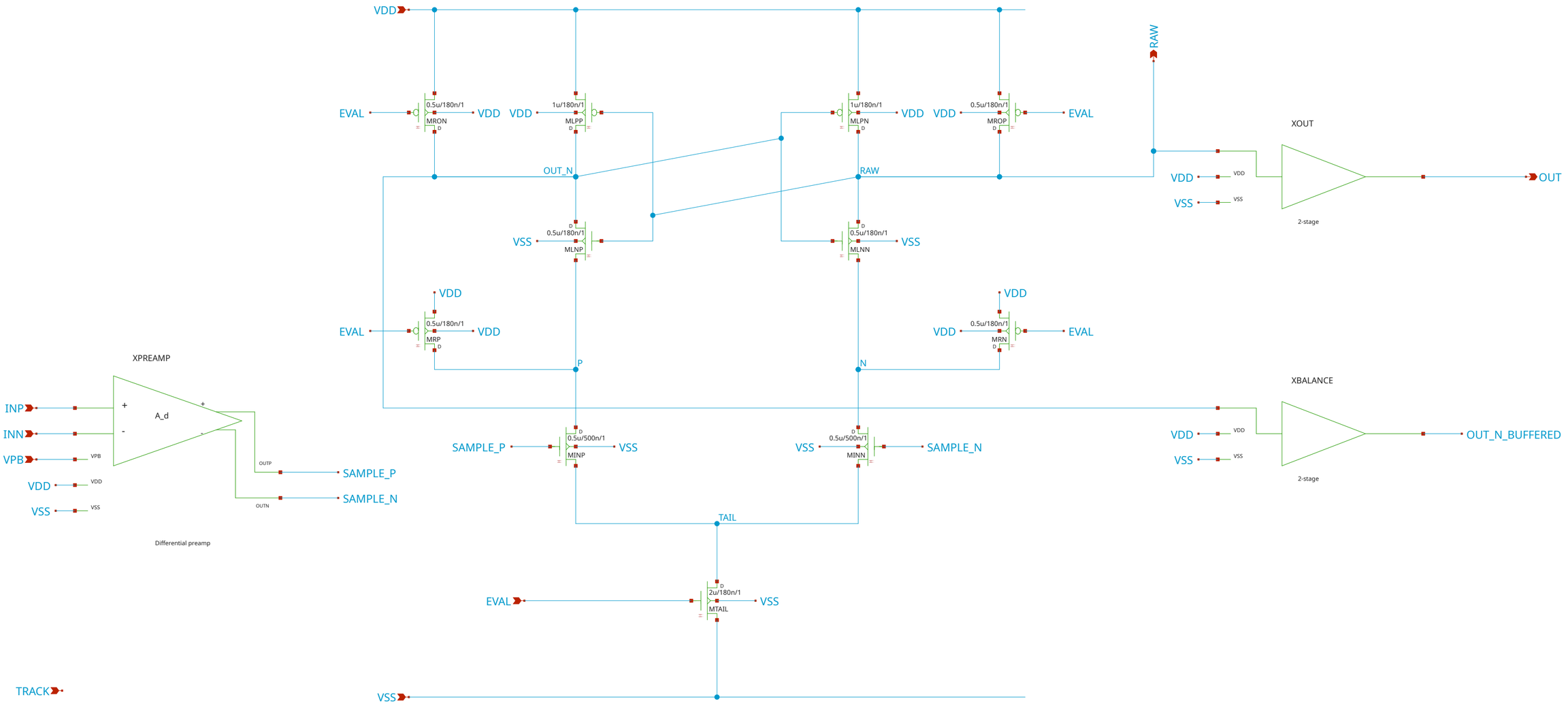


Bias conversion / mirror

Differential gain + physical low-pass capacitors

Preamplified StrongARM comparator | selected dynamic path

Continuous preamplifier at left; symmetric regeneration, clocked tail and explicit precharge at right. TRACK is retained as an unused interface pin.



Differential preamp

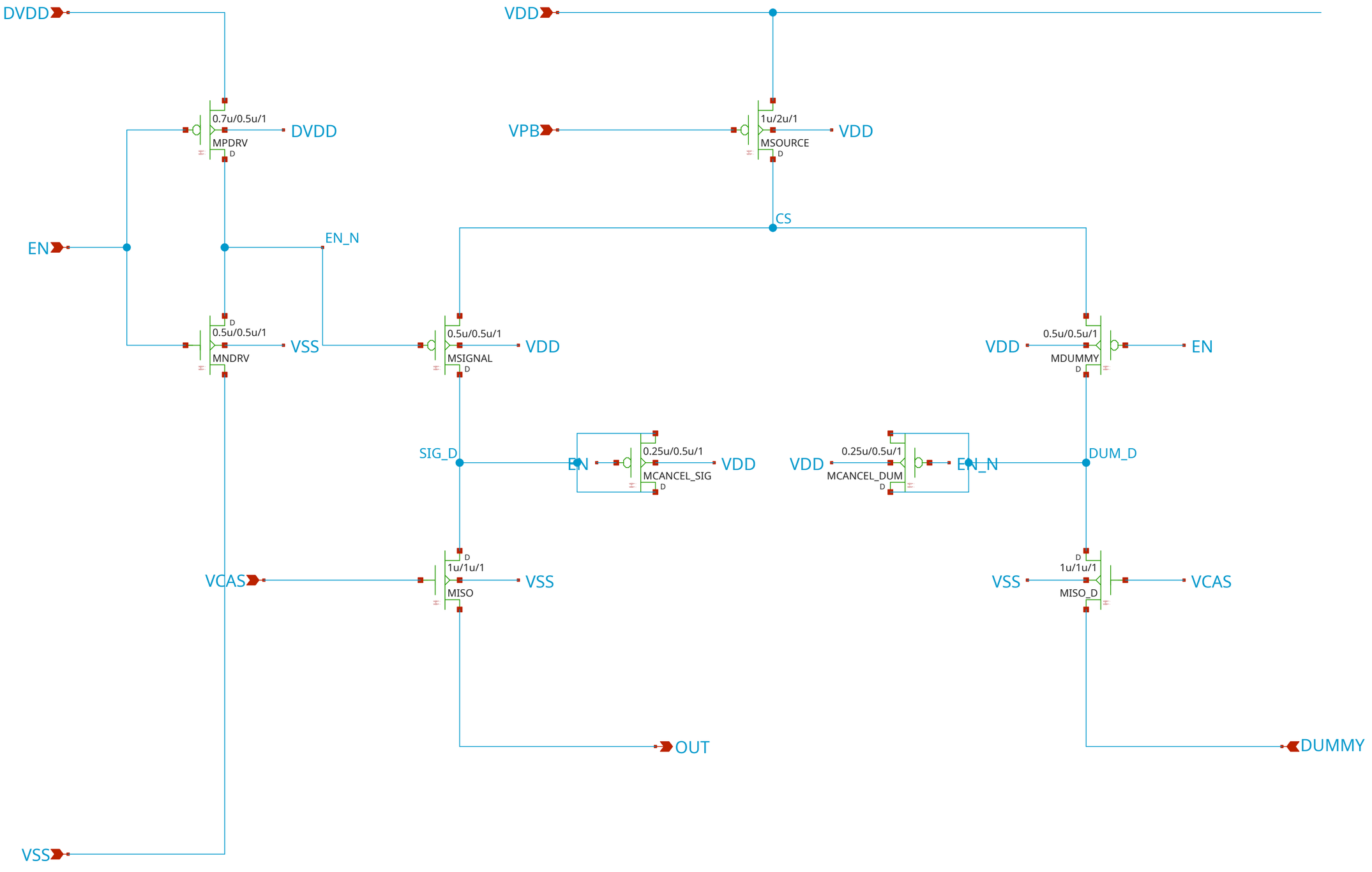
Preamp isolates input from regeneration; output filters are inside XPREAMP.

EXPLICIT C_LATCH LOADS | connected by named regenerative nodes; buffer and precharge devices retained



Calibrated replenishment | paired half-width PMOS dummies

Continuous source current is steered between SUM and a matched dummy branch. Common-gate devices isolate steering nodes from the outputs.



EN complement driver

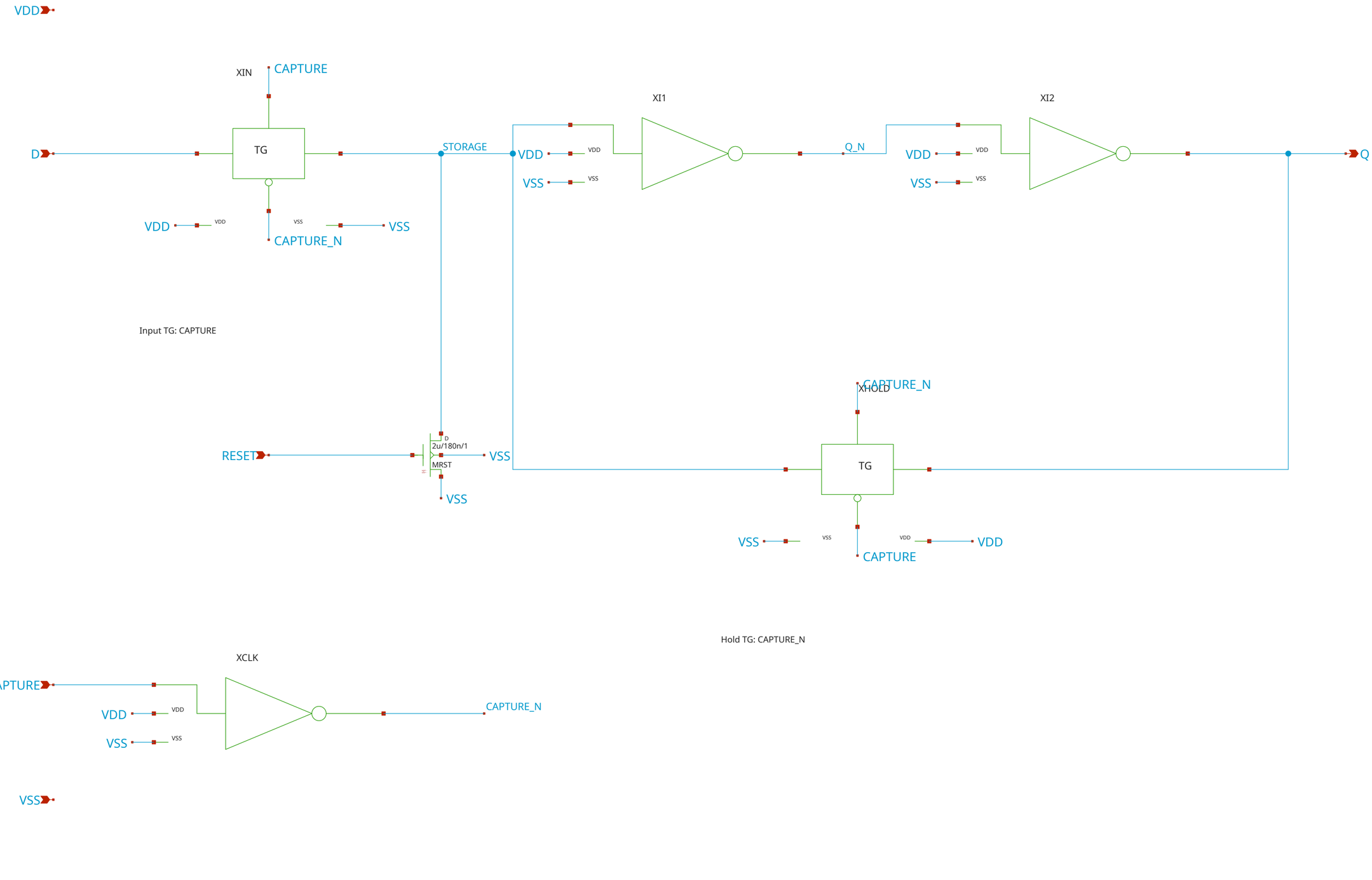
Signal branch

Dummy branch

Cancellation MOS D/S are visibly shorted at each internal steering drain. Gates are opposite to their main switch.

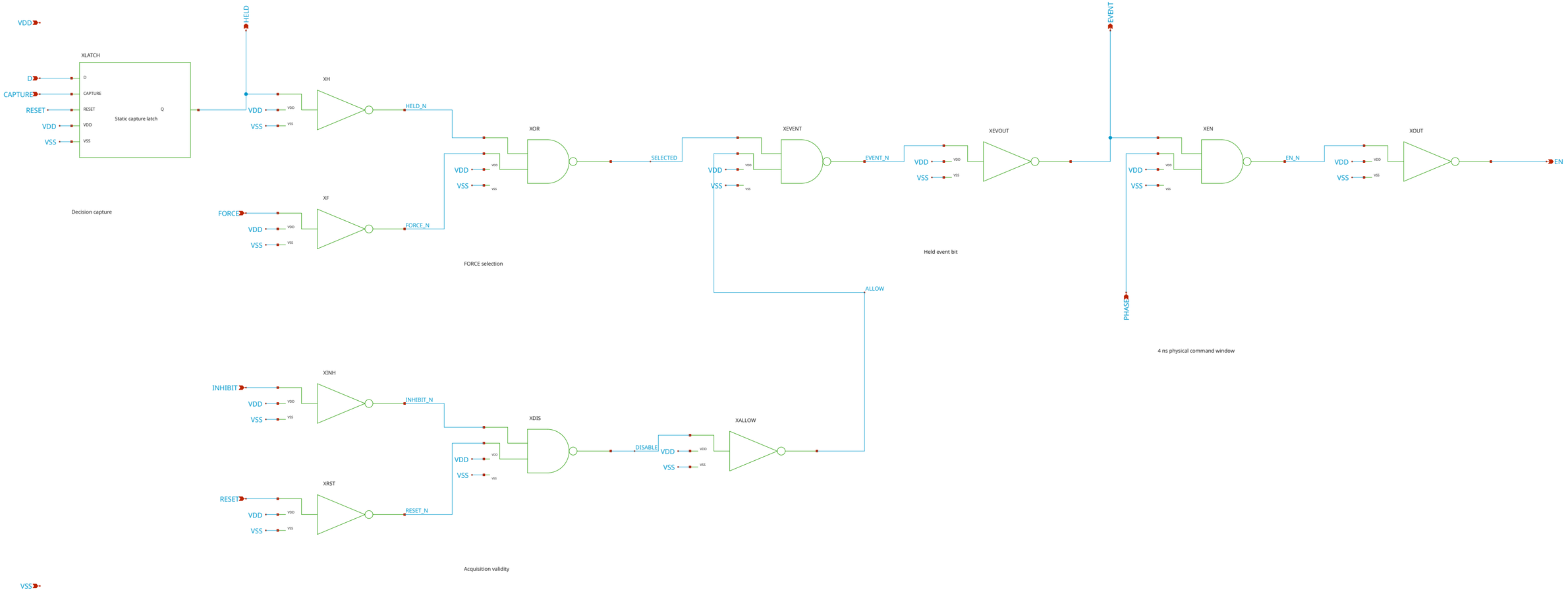
Static capture latch | transparent high

Forward data path above; complementary transmission-gate feedback below. RESET clamps STORAGE, without touching Cf.



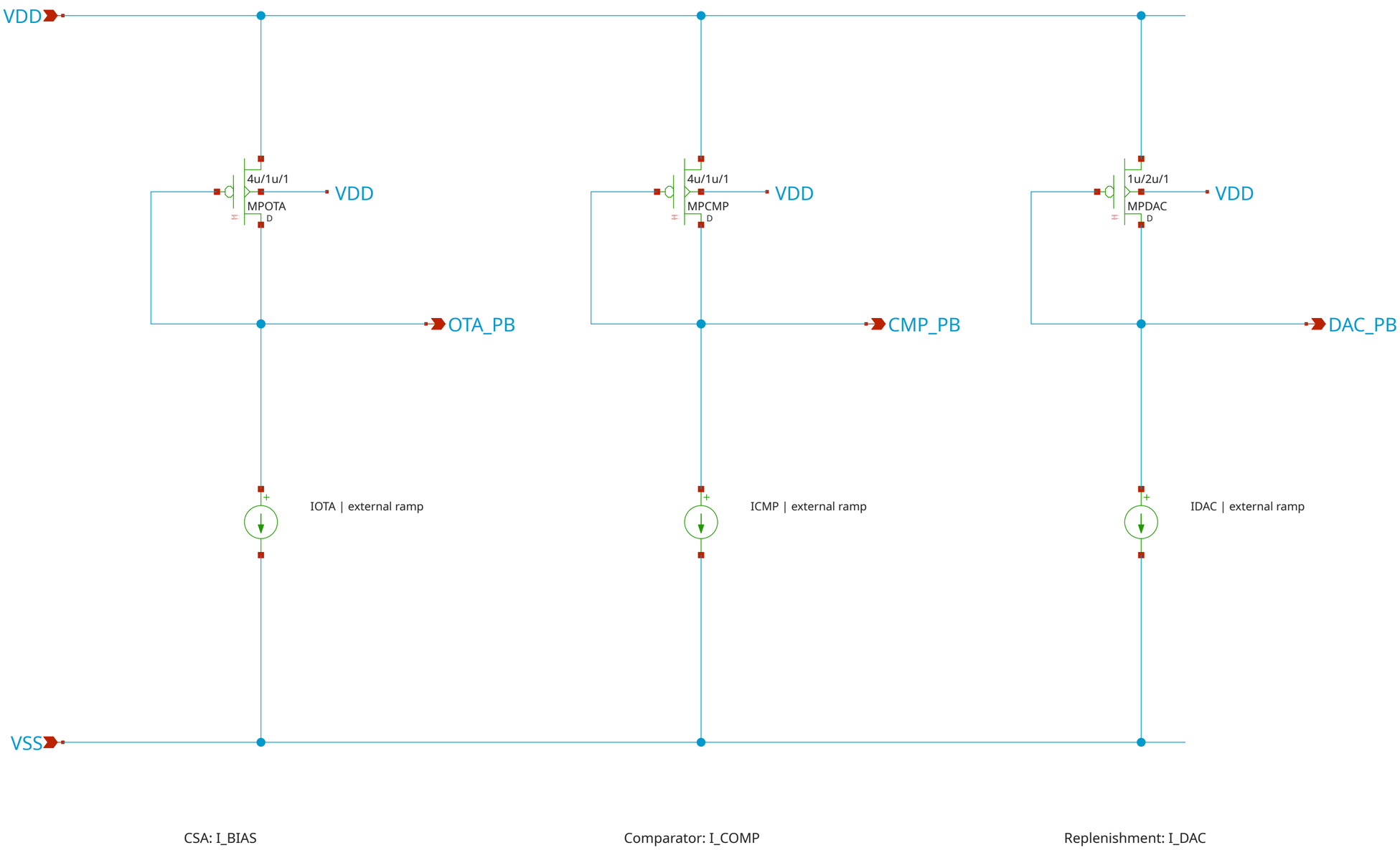
Captured replenishment control

EVENT = (HELD or FORCE) and not (INHIBIT or RESET). EN = EVENT and PHASE. Direct wires follow each implemented logic stage.



External-seed bias mirrors | explicit startup interface

Three diode-connected PMOS references. Current sources are external test references, with unchanged startup ramps; a shared generator is future work.

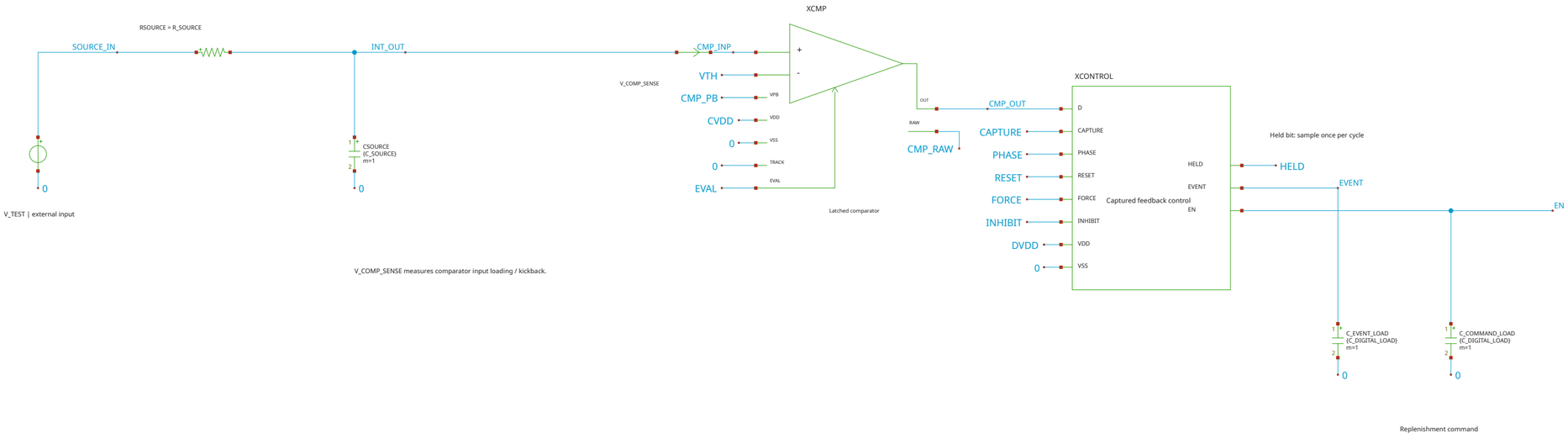


BIAS_INITIAL=1: constant seeds. BIAS_INITIAL=0: T_BIAS_READY / T_BIAS_RISE startup ramp.

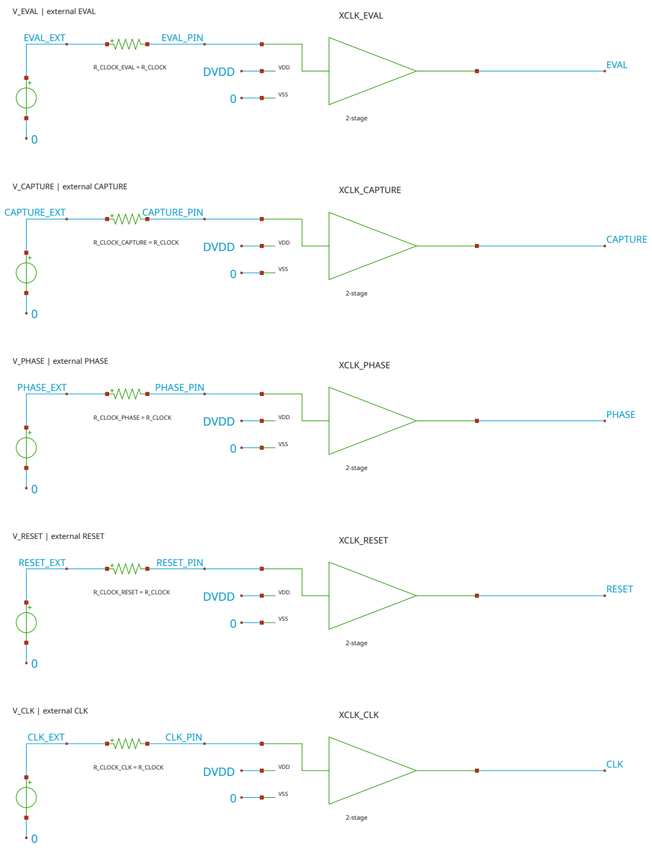
External TSMC180 model selector
Physical temperature = 298.15 K
Corner = TT

Comparator timing / kickback testbench

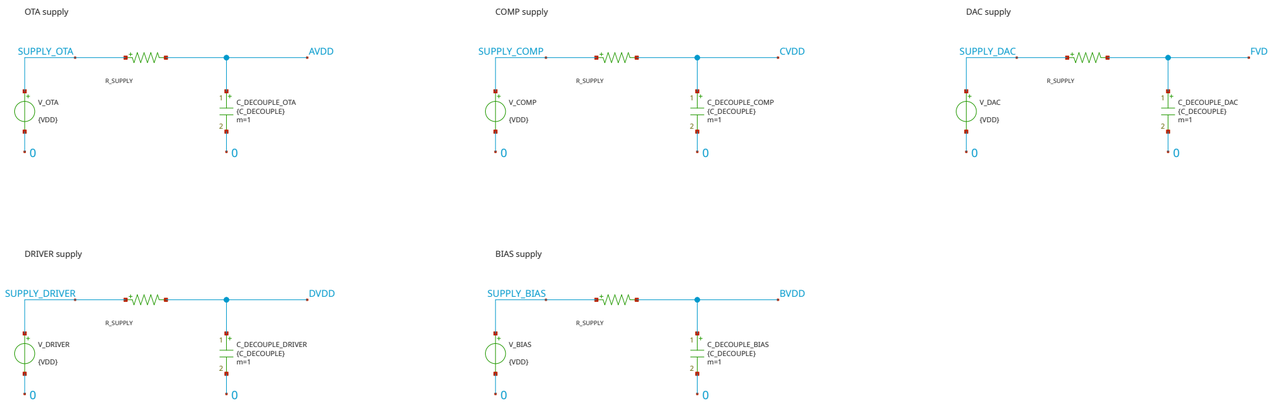
Finite input impedance -> comparator -> physical capture and feedback-window control.



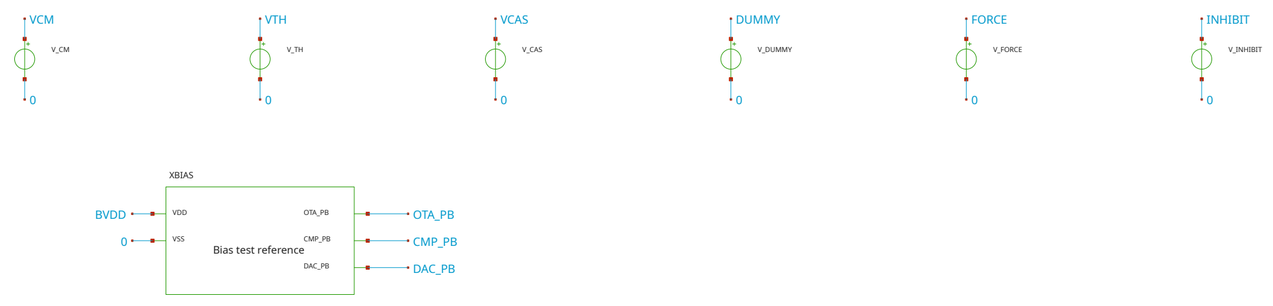
EXTERNAL TIMING + IMPLEMENTED LOCAL BUFFERS



FIVE SOURCE-POWER DOMAINS | SERIES FEED + DECOUPLING



IDEAL EXTERNAL REFERENCES / TEST CONTROLS



Diode bias interfaces; external current seeds remain explicit.

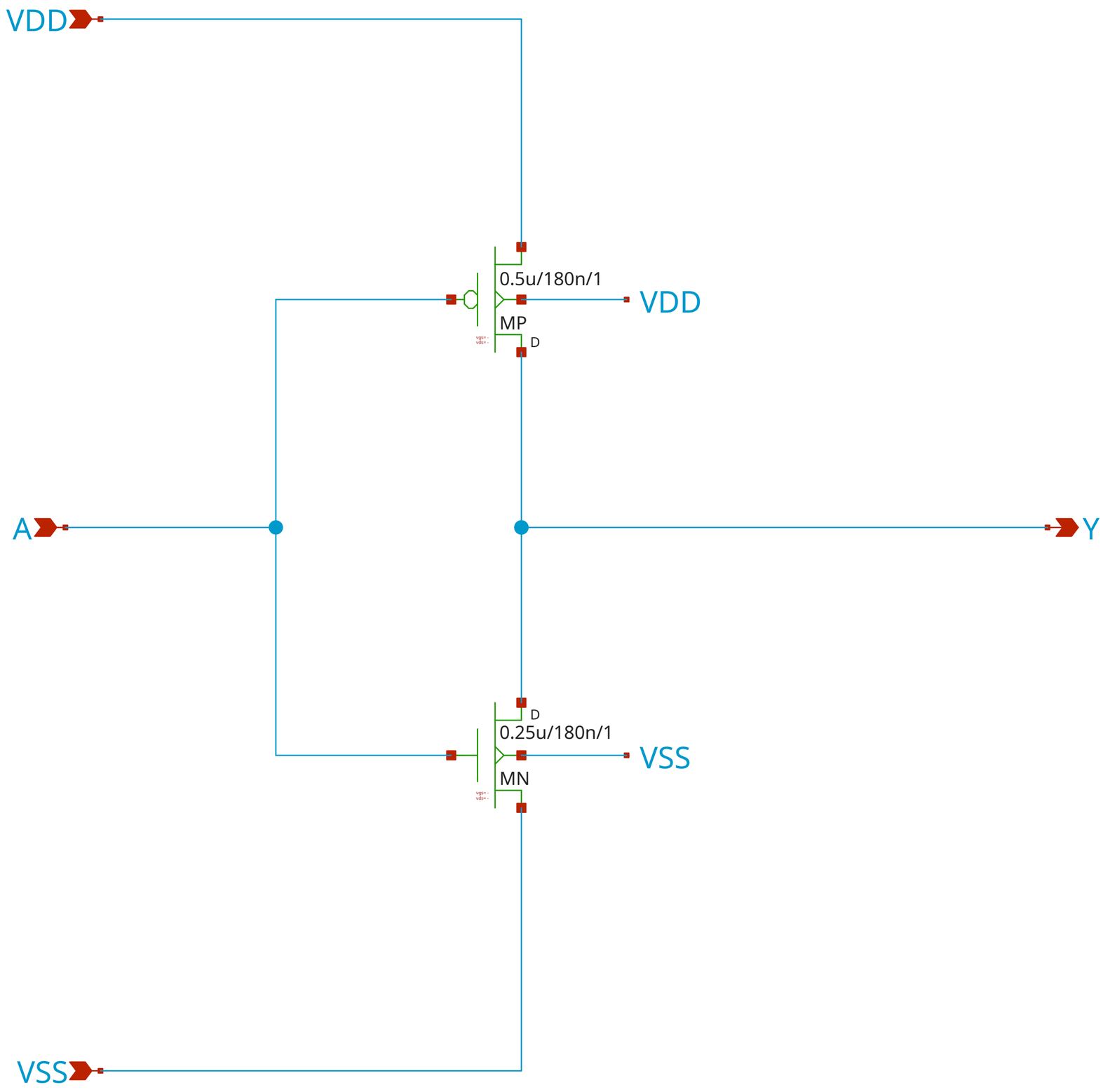
PARAMS | select the code block and press q for the complete unchanged simulator defaults.

GUI defaults: 1 MHz; C_IN=3 pF; Cf=50 fF; 1.8 V. Qualification recipes supply frozen trims and the selected late phases.

Every source, current probe, receiver load and power-feed element is retained. Colored net names connect support interfaces across panels.

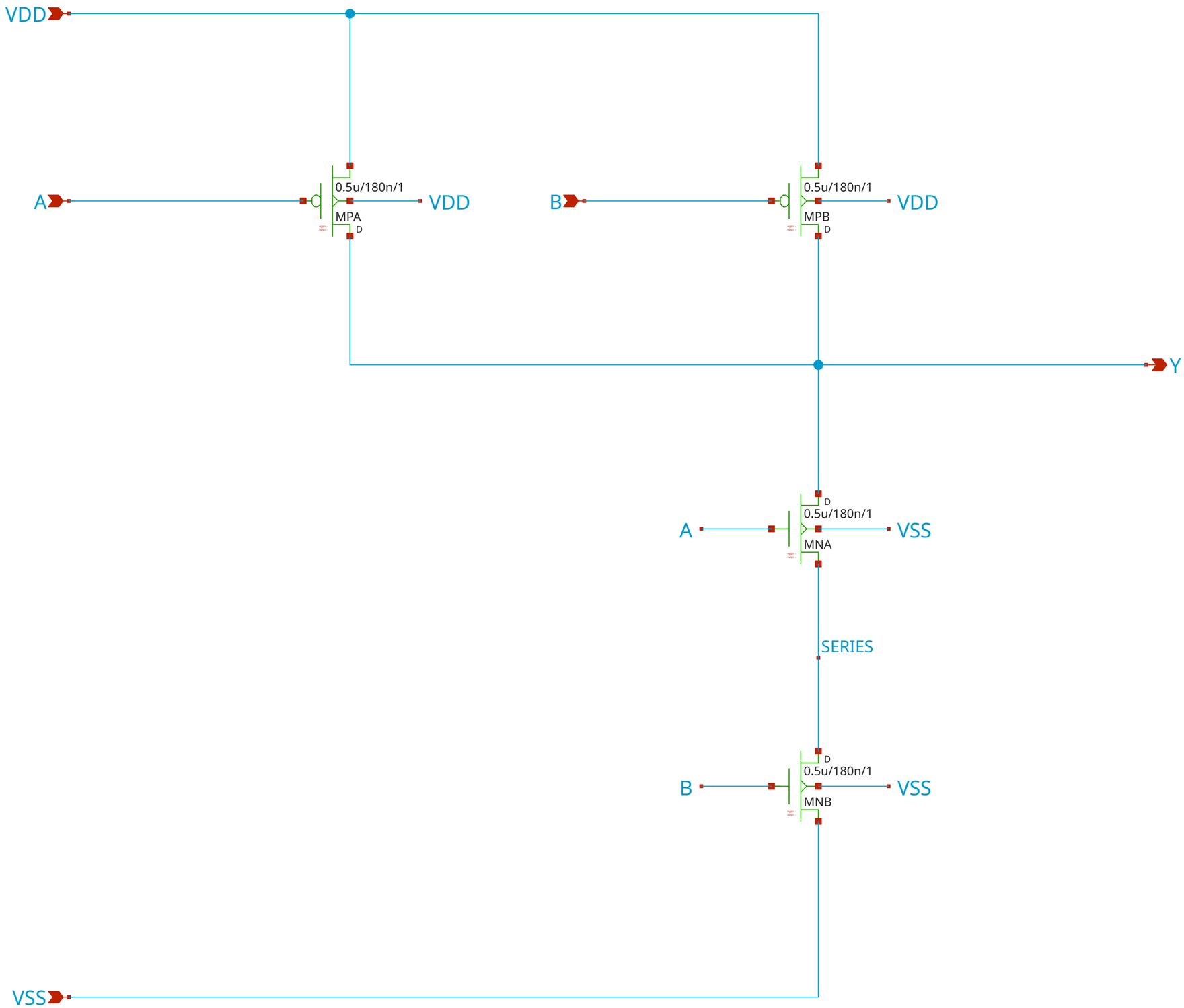
CMOS inverter

Complementary pull-up / pull-down; common input gate and common output drain.



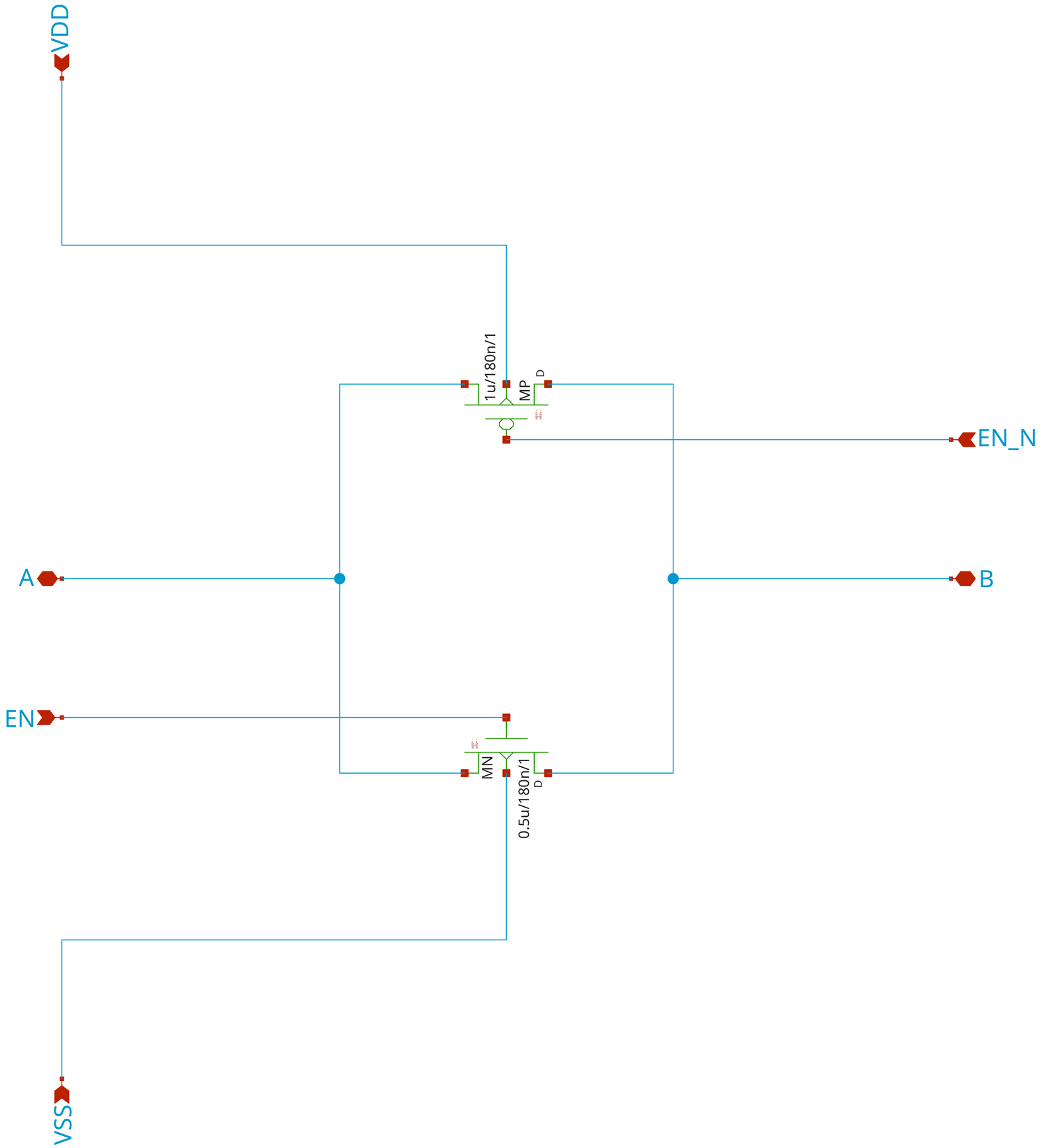
Two-input CMOS NAND

Parallel PMOS pull-up; series NMOS pull-down. Device names and dimensions are unchanged.



CMOS transmission gate

Parallel PMOS / NMOS channel; complementary enables. A and B remain bidirectional.



Local clock buffer

Two transistor-level CMOS inverters; the external clock and both power rails remain explicit.

